How Pruning Criteria Shape Post-RLVR Recovery in Reasoning Language Models

Anonymous ACL submission

Abstract

Structured pruning of reasoning Large Language Models (LLMs) is typically followed by Reinforcement Learning with Verifiable Rewards (RLVR) to recover reasoning ability. The standard pipeline treats the pruning decision as independent of the downstream RL alignment stage. We demonstrate that this independence assumption breaks down: recovery capacity is shaped at pruning time, not during subsequent fine-tuning. Through a controlled comparison of four structural pruning criteria under identical Group Relative Policy Optimization (GRPO) training, we observe consistent CKA-only advantages on DeepSeek-R1-Distill-Qwen-7B (GRPO recovery, 3-seed) and Qwen2.5-Math-7B (SFT recovery). CKA-only pruning, which removes only functionally redundant structures measured by Centered Kernel Alignment (CKA), achieves active learning with peak reward 0.354. Random pruning stagnates (peak 0.190). Standard importance-only pruning exhibits borderline collapse (peak 0.081). Recovery-Aware Structured Pruning (RASP) collapses permanently (peak 0.061). We trace the mechanism to preserved representational diversity: CKA similarity analysis shows that CKA-only pruning produces the lowest dense fidelity (μ=0.62 diagonal CKA) yet recovers best, while importance-based methods cluster near μ=0.90 and collapse. Gradient importance scoring consistently and significantly underperforms CKA-based scoring across our experimental conditions ($p<0.001$), as it removes structures that are representationally unique. GRPO reward gains do not translate to accuracy improvements across four benchmarks (GSM8K, MATH-500, HumanEval, MBPP; all GRPO deltas < 2 pp), while the recovery capacity gradient persists, confirming recovery capacity as the appropriate diagnostic for pruned model quality. The accuracy gap between CKA-only and Standard is robust across 3 GRPO seeds (4.55 pp $\pm$ 0.33). Our results demonstrate that pruning criterion selection is an important factor in post-pruning RL trainability, and that CKA-based redundancy scoring consistently outperforms gradient importance for RL-intended compression.

1 Introduction

The choice of which structures to prune from a reasoning LLM shapes whether the model sustains reinforcement learning alignment. Current compression pipelines treat this choice as independent of downstream training: structured pruning removes entire attention heads or MLP channels to reduce memory and latency (Ma et al., 2023; Wang et al., 2026), Supervised Fine-Tuning (SFT) recovers task accuracy, and Reinforcement Learning with Verifiable Rewards (RLVR) restores reasoning quality (DeepSeek-AI, 2025; Shao et al., 2024). Each stage is optimized in isolation. We demonstrate that this separation fails: the pruning criterion significantly influences what we term **recovery capacity**, the model's ability to sustain RL-based recovery without collapsing into degenerate outputs.

The consequences of ignoring this coupling are severe: GRPO training on pruned models can trigger complete collapse, with gradient norms approaching zero, repetitive outputs, and monotonically degrading reward. Whether collapse occurs depends entirely on which structures were removed. Reward itself is an unreliable diagnostic: on unpruned DeepSeek-R1-Distill-Qwen-7B (DeepSeek-AI, 2025), 100 steps of Group Relative Policy Optimization (GRPO; Shao et al., 2024) raise mean reward from 0 to 0.669 while greedy accuracy on GSM8K (Cobbe et al., 2021) remains exactly 81.27%.[1] This reward-accuracy decoupling motivates our focus on recovery capacity as the appropriate diagnostic for post-pruning RL trainability.

[1]The unpruned model achieves 90.83% without fine-tuning. Applying the same SFT recovery protocol designed for pruned models reduces accuracy to 81.27%, consistent with catastrophic forgetting when further fine-tuning an already well-optimized model.

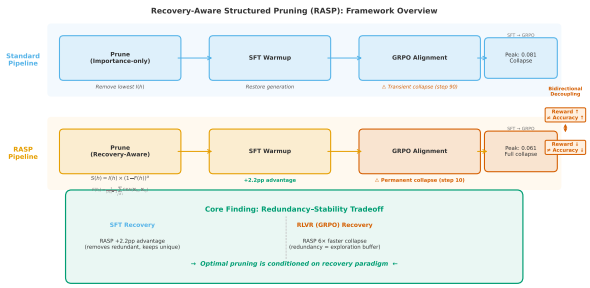


Figure 1: Controlled comparison framework. All four pruning criteria share the same structural analysis (Taylor importance + CKA overlap) and identical recovery pipeline (SFT warmup → GRPO). Despite identical training, the criteria produce divergent outcomes: CKA-only achieves active learning while Standard and RASP collapse.

Ji et al. (2025) claim that RLVR recovers pruned reasoning models to 95%+ of original performance, but their evaluation uses a single pruning method and reports only final-checkpoint accuracy without examining training dynamics.

Existing structured pruning methods for reasoning LLMs (Wang et al., 2026, 2025; Ma et al., 2023; Kolawole et al., 2025; He et al., 2025; Liu et al., 2025b) optimize for post-pruning quality without studying downstream alignment; none compare multiple pruning criteria under RL or study how pruning affects RL training dynamics. Matsutani et al. (2026) show that RL and SFT impose fundamentally different demands on model representations (RL compresses solution paths while SFT expands them), suggesting that pruning decisions should substantially affect RL trainability, yet no prior work has tested this hypothesis.

We present a controlled comparison of four structural pruning criteria under identical GRPO training on DeepSeek-R1-Distill-Qwen-7B at 20% structured sparsity, with cross-architecture validation on Qwen2.5-Math-7B (Figure 1). The four criteria span a spectrum of representational diversity preservation: **CKA-only** scoring removes structures with the highest pairwise Centered Kernel Alignment (CKA; Kornblith et al., 2019) overlap, preserving all representationally unique structures; **Random** scoring provides an unbiased baseline; **Standard** importance-only scoring retains structures with the highest gradient-based importance (Ma et al., 2023); and **Recovery-Aware Structured Pruning** (RASP) combines importance with CKA overlap in a composite score. Under identical GRPO configuration, these four methods reveal a **recovery capacity gradient**. CKA-only pruning achieves *active learning*: peak reward reaches 0.354 (4.4× Standard). Random pruning stagnates (reward oscillates 0.14–0.19). Standard pruning exhibits borderline collapse (peak 0.081). RASP collapses permanently (peak 0.061, 85% of completions yield zero reward variance). The ordering, CKA-only ≫ Random > Standard > RASP, follows the degree of preserved representational diversity. Gradient importance scoring, the standard criterion for LLM pruning, removes structures scored "unimportant" by gradient metrics that are in fact representationally unique, reducing the exploration diversity available for RL training in our setting. A depth pruning baseline (ShortGPT; Men et al., 2024) is the only method where GRPO actively degrades accuracy (-1.36 pp), confirming that pruning granularity interacts with RL trainability.

Our contributions are:

1. We discover a **recovery capacity gradient** across four pruning criteria under controlled GRPO training: CKA-only enables active learning (peak reward 4.4× Standard), Random stagnates, and Standard/RASP collapse. Extended training (300 steps) confirms this gap is structural, not a matter of training budget (§4.2, §4.5).
2. We identify **preserved representational diversity** as a plausible mechanism: CKA similarity analysis shows that the best-recovering method (CKA-only, μ=0.62 diagonal CKA) diverges most from the dense model, while collapsing methods preserve dense fidelity ($\mu \geq 0.88$). Cross-architecture experiments on Qwen2.5-Math-7B and cross-scale experiments on 1.5B provide initial evidence for the generality of the ordering (§4.5).

2 Related Work

Structured pruning of LLMs. LLM-Pruner (Ma et al., 2023) introduced dependency-aware structural pruning for transformers; Bonsai (Kolawole et al., 2025) achieves competitive results using only forward passes; GRASPrune (Wang et al., 2026) advances the state of the art with global gating that jointly prunes FFN channels and KV head groups. ShortGPT (Men et al., 2024) and SliceGPT (Ashkboos et al., 2024) target depth pruning, while Wanda (Sun et al., 2024) addresses unstructured sparsity; Monjur et al. (2026) revisit the structured-

vs-unstructured trade-off under test-time scaling. For reasoning models, RESP (Wang et al., 2025) uses gradient-based importance on self-generated reasoning traces, RAC (Lucas et al., 2025) reconstructs chain-of-thought during pruning, SSGR (Xiang et al., 2025) uses self-generated calibration data, and Ding et al. (2026) find width pruning more robust than depth pruning for reasoning but do not study RLVR recovery. A parallel line recognizes that functional structure should guide pruning: FANG (Yu et al., 2025) groups FFN neurons by activation clusters, FuncNet (Liu et al., 2025a) decomposes LLMs via ICA, and MPruner (Hu et al., 2024) uses inter-layer CKA (Kornblith et al., 2019) for depth pruning in CNNs. Our work uses *intra-layer* CKA between individual structures at a finer granularity, optimizing for RL recovery capacity rather than immediate accuracy preservation.

Pruning-aware recovery. Ji et al. (2025) demonstrate that RLVR restores pruned reasoning models to 95%+ accuracy, significantly outperforming SFT, but test only one pruning method without studying whether the pruning criterion affects recovery. P^2 Law (Chen et al., 2025) establishes scaling laws for post-training loss after pruning. PASER (He et al., 2025) optimizes recovery data selection, PAT (Liu et al., 2025b) jointly considers pruning and fine-tuning objectives, RestoreLCC (Feng et al., 2025) contrastively probes critical attention heads for post-hoc compensation, Zhao et al. (2026) adopt iterative prune-tune loops for gradual compression, and OPSD (Kim and Lee, 2026) compresses reasoning models *after* RLVR (the reverse of our pipeline). These approaches optimize complementary axes (data, training, compensation, schedule); our work isolates which *pruning criterion* enables RL-based recovery. The Lottery Ticket Hypothesis (Frankle and Carbin, 2019) asks which subnetworks train effectively from random initialization; our recovery capacity gradient extends this to *paradigm-conditioned trainability*, where the optimal subnetwork depends on the recovery paradigm (RLVR vs. SFT).

RLVR training dynamics. Matsutani et al. (2026) show that RLVR *compresses* incorrect reasoning trajectories via global exploration, while SFT *expands* correct ones via local imitation. This asymmetry motivates our finding: gradient importance scoring protects structures that RL would compress, while removing the diverse representations RL exploration requires. RL training instability has been studied in the RLHF context: Gao et al. (2023) characterize reward overoptimization, Dong et al. (2025) document repetitive output collapse during extended RL training, and Shah et al. (2025) analyze compounding bias from reward estimation. Our work connects this instability literature to structural pruning, showing that the pruning criterion shapes susceptibility to RL collapse.

3 Method

We design a controlled comparison framework to isolate the effect of pruning criterion on post-RLVR recovery capacity. The framework consists of three components: (1) shared structural analysis that computes importance and functional overlap scores for all prunable structures, (2) four scoring criteria that combine these scores differently to select structures for removal, and (3) an identical two-stage recovery pipeline applied to all pruned models. By varying only the scoring criterion while holding all other factors constant, we can attribute differences in GRPO training dynamics directly to the pruning decision.

3.1 Preliminaries and Notation

Structured pruning. Given a pretrained transformer $\mathcal{M}$ with L layers, structured pruning removes entire computational units (attention heads or MLP channels) to reduce model size while maintaining hardware-friendly dense computation. We denote the set of prunable structures as $\mathcal{H} = \{h_1, h_2, \ldots, h_N\}$.

Grouped Query Attention (GQA). In GQA architectures (Ainslie et al., 2023), n_Q query heads share n_{KV} key-value heads, with each KV head serving a group of $G = n_Q/n_{KV}$ query heads. For DeepSeek-R1-Distill-Qwen-7B (n_Q=28, n_{KV}=4), $G = 7$, yielding 4 GQA groups per layer. We treat each GQA group (7 Q heads + 1 KV head) as a single prunable unit, since removing a KV head necessitates removing all associated Q heads.

RLVR and GRPO. Reinforcement Learning with Verifiable Rewards (RLVR) recovers pruned model capabilities using outcome-based rewards from tasks with verifiable answers. We use Group Relative Policy Optimization (GRPO; Shao et al., 2024) as the recovery algorithm.

3.2 Structural Analysis

All four pruning criteria share a common structural analysis stage that computes two scores per structure: *importance* $I(h)$ capturing the structure's contribution to reasoning, and *functional overlap* $F(h)$ measuring representational redundancy with other structures in the same layer.

Calibration data. We compute importance and overlap scores on chain-of-thought reasoning data from GSM8K (Cobbe et al., 2021) training examples. Given a calibration set $\mathcal{D}_{\text{cal}} = \{(x_j, y_j)\}_{j=1}^{M}$, we compute activations and gradients on the concatenated sequences (x_j, y_j).

Structural importance. For each structure h, we compute importance $I(h)$ via first-order Taylor expansion on the reasoning trace loss:

$$I(h) = \left| \sum_{(x,y)\in\mathcal{D}_{\text{cal}}} \mathbf{a}_h(x,y)^\top \frac{\partial \mathcal{L}(x,y)}{\partial \mathbf{a}_h} \right|, \quad (1)$$

where $\mathbf{a}_h$ denotes the activation of structure h and $\mathcal{L}$ is the language modeling loss over the reasoning trace. For a GQA group g containing Q heads $\{q_1, \ldots, q_G\}$ and KV head k, we aggregate: $I(g) = \sum_{i=1}^{G} I(q_i) + I(k)$.

Functional overlap via CKA. We measure functional overlap using linear Centered Kernel Alignment (CKA; Kornblith et al., 2019), which quantifies similarity between two sets of representations invariant to invertible linear transformations. Given activation matrices $\mathbf{X}, \mathbf{Y} \in \mathbb{R}^{n\times d}$ collected over n calibration examples, linear CKA is:

$$\text{CKA}(\mathbf{X}, \mathbf{Y}) = \frac{\text{HSIC}(\mathbf{X}, \mathbf{Y})}{\sqrt{\text{HSIC}(\mathbf{X}, \mathbf{X}) \cdot \text{HSIC}(\mathbf{Y}, \mathbf{Y})}}, \quad (2)$$

where $\text{HSIC}(\mathbf{X}, \mathbf{Y}) = \frac{1}{(n-1)^2} \text{tr}(\mathbf{KHLH})$, with $\mathbf{K} = \mathbf{X}\mathbf{X}^\top$, $\mathbf{L} = \mathbf{Y}\mathbf{Y}^\top$, and $\mathbf{H} = \mathbf{I}_n - \frac{1}{n}\mathbf{1}\mathbf{1}^\top$.

For a GQA group g_i in layer ℓ, we form its representation by concatenating its constituent Q head activations: $\mathbf{X}_{g_i} = [\mathbf{a}_{q_1}; \ldots; \mathbf{a}_{q_G}] \in \mathbb{R}^{n\times G d_h}$. The functional overlap of g_i is the mean pairwise CKA with other groups in the same layer:

$$F(g_i) = \frac{1}{|\mathcal{G}_\ell| - 1} \sum_{\substack{g_j \in \mathcal{G}_\ell \\ j \neq i}} \text{CKA}(\mathbf{X}_{g_i}, \mathbf{X}_{g_j}). \quad (3)$$

We apply the same intra-layer CKA overlap to MLP channel groups, partitioning each MLP intermediate dimension into equal-sized groups. Unlike prior work that uses inter-layer CKA to identify globally redundant layers for depth pruning (Hu et al., 2024; Lachance, 2022), we use *intra-layer* CKA between individual structures to measure per-structure functional redundancy. Computing all pairwise CKA scores adds negligible overhead: with at most 28 structures per layer and 28 layers, the total number of CKA evaluations is $O(L \cdot k^2)$ where $k \leq 28$, completing in under 2 minutes on a single GPU.

CKA reliability. Our intra-layer use of linear CKA between structures of identical dimensionality avoids the kernel width and layer width sensitivities identified by Davari et al. (2023); Klabunde et al. (2023); empirically, structure rankings are stable across calibration sizes (Spearman $\rho \geq 0.964$; Appendix H).

3.3 Pruning Scoring Criteria

We compare four scoring criteria that combine importance $I(h)$ and functional overlap $F(h)$ in different ways. All four methods use the same per-layer capacity floor ($\geq$40% of MLP groups per layer must be retained, chosen to prevent any single layer from becoming a throughput bottleneck) and global ranking procedure: structures are sorted by score, and the lowest-scoring structures are pruned up to the target sparsity ratio, skipping any candidate whose removal would violate the layer floor.

Standard (importance-only). Following LLM-Pruner (Ma et al., 2023), structures are ranked solely by importance:

$$S_{\text{std}}(h) = I(h). \quad (4)$$

This is the industry-standard criterion that retains gradient-important structures and removes "unimportant" ones.

RASP (recovery-aware composite). The recovery-aware score combines importance and functional overlap:

$$S_{\text{rasp}}(h) = I(h) \times \big(1 - F(h)\big)^{\alpha}, \quad (5)$$

where $\alpha \geq 0$ controls the strength of the overlap penalty (α=0 reduces to Standard). Structures with high importance *and* low overlap receive the highest scores and are retained; structures with high overlap are penalized and pruned preferentially, regardless of their importance.

CKA-only (overlap-only). CKA-only scoring ignores gradient importance entirely, ranking structures by functional overlap alone. Structures with the highest mean pairwise CKA similarity to other structures in their layer are removed first:

$$S_{\text{cka}}(h) = 1 - F(h). \tag{6}$$

This criterion preserves all representationally unique structures regardless of their gradient importance, removing only those whose function is redundant with retained structures. S_{cka} corresponds to the limit $\alpha \to \infty$ in Eq. 5.

Random. Structures are selected uniformly at random for removal, subject to the per-layer capacity floor. This baseline controls for the effect of any informed scoring criterion.

The four criteria span a spectrum from pure gradient importance (Standard) to pure representational diversity preservation (CKA-only), with RASP as a weighted interpolation and Random as an uninformed control. The key distinction is that CKA-only protects *functionally unique* structures regardless of gradient contribution: structures deemed "unimportant" by gradient metrics may provide non-redundant computational pathways essential for RL exploration (Matsutani et al., 2026).

3.4 Two-Stage Recovery

After pruning, all four models undergo an identical two-stage recovery pipeline. Any differences in GRPO training dynamics are therefore attributable to the pruning criterion alone.

Stage 1: SFT warmup. Structured pruning disrupts the model's ability to generate end-of-sequence (EOS) tokens: in sampling mode, pruned models produce repetitive outputs without termination. Since GRPO computes rewards only from completed generations, this EOS collapse makes direct RLVR recovery impossible. We restore generation competence through a short SFT phase on chain-of-thought reasoning demonstrations with teacher forcing. This stage is intentionally brief: its purpose is to restore the minimal capabilities (EOS generation, format compliance) needed for GRPO, not to recover full reasoning ability.

Stage 2: GRPO alignment. With generation competence restored, we apply GRPO on mathematical reasoning problems with binary reward (r=1 if the final answer matches the ground truth, r=0 otherwise). RLVR is preferred over extended SFT for reasoning recovery because of their fundamentally different training dynamics: Matsutani et al. (2026) show that RLVR *compresses* incorrect reasoning trajectories while SFT *expands* correct ones. For a pruned model, RLVR can globally explore and redistribute functionality across remaining structures, while SFT is limited to imitating fixed trajectories through a reduced architecture. The GRPO configuration (learning rate, KL coefficient, sampling temperature, loss formulation) is held constant across all four pruned models (Section 4.1).

3.5 Recovery Capacity Diagnostics

We define **recovery capacity** as the model's ability to benefit from post-pruning recovery training, operationalized in our GRPO experiments as peak mean reward over the training horizon (Table 1). We track four diagnostic measures—mean reward, zero-variance batch fraction (f_0), gradient norm, and KL divergence—to distinguish *active learning* (rising reward, low f_0), *stagnation* (flat reward), and *collapse* (declining reward, high f_0, vanishing gradients).

The complete pruning comparison procedure is given in Algorithm 1 (Appendix B).

4 Experiments

4.1 Experimental Setup

Base model and evaluation. We evaluate on DeepSeek-R1-Distill-Qwen-7B (DeepSeek-AI, 2025), a 7B-parameter reasoning model built on the Qwen2.5 architecture (Yang et al., 2024) with GQA (Ainslie et al., 2023) (architecture details in Section 3.1). We evaluate on GSM8K (Cobbe et al., 2021), a benchmark of 1,319 grade-school math problems requiring multi-step arithmetic reasoning. We report exact-match accuracy: the final numerical answer is extracted from the model's generation and compared against the ground truth. For rapid iteration during development, we use a 100-problem subset (N=100); final results use the full test set (N=1,319). The original unpruned model achieves 90.83% on GSM8K (greedy decoding).

Pruning methods. We compare four pruning criteria at 20% structured sparsity, all removing the same number of MLP channel groups (200 of 784 across 28 layers, ∼1.45B parameters) with identical per-layer capacity floors (≥40% retention per layer):

1. **CKA-only**: structures ranked by functional overlap $F(h)$ alone (Eq. 3); the most redundant groups are removed regardless of gradient importance.
2. **Random**: uniform random selection among eligible groups.
3. **Standard**: Taylor-expansion importance $I(h)$ alone (Eq. 1); the least important groups are removed, following LLM-Pruner (Ma et al., 2023).
4. **RASP** (α=1.0): the composite score $S(h) = I(h) \times (1 - F(h))^{\alpha}$ (Eq. 5), removing structures that are both low-importance and functionally redundant.

Recovery pipeline and hardware. All four methods undergo matched two-stage recovery. Stage 1 (SFT warmup): 2 epochs on 7,473 GSM8K chain-of-thought examples with teacher forcing (37 minutes), restoring EOS generation and basic format compliance. Stage 2 (GRPO): Group Relative Policy Optimization (Shao et al., 2024) with learning rate 1×10^{-6}, maximum completion length 1,024 tokens, KL penalty coefficient β=0.01, sampling temperature 0.4, DAPO-style clipped loss, and binary reward (r=1 if correct, r=0 otherwise), trained for 100 steps on 1,000 GSM8K training problems. All methods share identical GRPO hyperparameters, including `reward_desert_patience`=50. Seeds vary at the GRPO stage; pruning calibration and SFT use identical configurations across all methods. We repeat GRPO training with 3 random seeds for CKA-only and Standard (Table 2). CKA-only achieves 64.39% ± 0.19 pp (std across seeds), Standard 59.84% ± 0.38 pp; the 4.55 pp gap is robust across seeds (p=0.0009), confirming that recovery capacity differences are structural rather than stochastic. All experiments run on 2× NVIDIA RTX PRO 6000 GPUs (94 GB VRAM each); pruning variants are trained in parallel on separate GPUs under identical wall-clock conditions.

4.2 Recovery Capacity Gradient

The central finding of this work is that the pruning criterion can significantly influence post-RLVR recovery capacity in our experimental setting. Under controlled GRPO training with identical hyperparameters, the four pruning methods produce qualitatively distinct training dynamics that form a clear recovery capacity gradient: CKA-only ≫ Random ≫ Standard > RASP. Table 1 summarizes the key metrics, and Figure 2 shows the reward trajectories over 100 GRPO steps.

Table 1: Recovery capacity gradient: GRPO dynamics for four pruning criteria at 20% sparsity under identical configuration (lr 1×10^{-6}, β=0.01, T=0.4, 100 steps). Peak reward = best across 100 training steps. f_0: fraction of training batches where all sampled completions receive identical reward (lower = healthier exploration). Best in **bold**, second-best underlined.

Method	Peak Rew. ↑	Mean Rew. ↑	Min f_0 ↓	Behavior
CKA-only	**0.354**	**0.285**	**0.125**	Active learning
Random‡	<u>0.190</u>	<u>0.164</u>	<u>0.300</u>	Stagnation
Standard	0.081	0.050	0.500	Borderline collapse
RASP‡	0.061	0.031	0.675	Full collapse

‡Single seed (N=1). CKA-only and Standard validated with 3 seeds (Table 2).

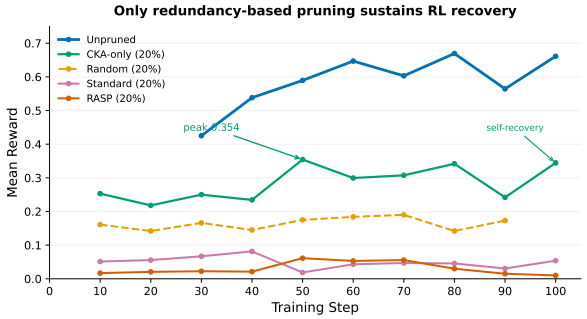


Figure 2: GRPO reward trajectories for four pruning criteria and the unpruned model under identical training configuration (β=0.01, T=0.4, 100 steps). CKA-only reaches 0.354 peak reward with an observed recovery event at step 90. Random oscillates between 0.14 and 0.19 (training timed out at step 97; 9 steps shown). Standard peaks at 0.081 before borderline collapse. RASP peaks at 0.061 (step 50) and declines to 0.010 by step 100.

CKA-only pruning is the only method that produces active learning under GRPO. Reward reaches a peak of 0.354 (mean 0.285) with f_0 as low as 0.125, including a recovery event at step 90 where reward rebounds from a transient perturbation. Random pruning produces stagnation: reward oscillates between 0.142 and 0.190 with f_0 near 0.33.

Standard pruning exhibits borderline collapse (Figure 3): reward peaks at 0.081 (step 40) then oscillates near 0.05, with $f_0 \geq 0.500$ throughout—more than half of batches produce no learning signal. RASP produces full collapse: reward peaks at 0.061 (step 50) then declines to 0.010 by step 100, with $f_0 > 0.675$ and clipped ratio > 90%.

The 5.8× peak reward gap (CKA-only 0.354 vs RASP 0.061) arises from models pruned to identical

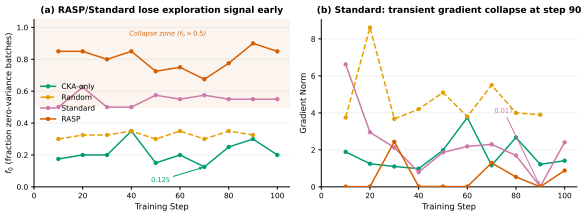


Figure 3: Training dynamics diagnostics. (a) f_0 (fraction of zero-variance batches): CKA-only stays below 0.35; Standard and RASP remain in the collapse zone ($f_0 > 0.5$) throughout training. (b) Gradient norm: Standard exhibits a transient minimum of 0.017 at step 90 before recovering to 2.41 at step 100; RASP gradient norms remain predominantly near zero.

Table 2: GRPO responsiveness on GSM8K (N=1,319, greedy decoding). All pruned models use the controlled v5 GRPO configuration (identical hyperparameters including `reward_desert_patience`=50), evaluated at step 100. CKA-only and Standard report mean ± std over 3 GRPO seeds; RASP uses a single seed. Best in **bold**, second-best underlined.

Model	SFT Acc. (%)	GRPO Acc. (%)	Δ (pp)	Peak Rew.
Unpruned	**81.27**	**81.27**	0.00	**0.669**
CKA-only 20%	63.91	64.39±0.19	+0.48	0.354
Standard 20%	59.29	59.84±0.38	+0.55	0.081
RASP 20%	61.49	60.88	−0.61	0.061
Random 20%‡	–	–	–	0.190
ShortGPT 20%†	44.73	43.37	−1.36	0.000

†Depth pruning baseline (Men et al., 2024); removes entire layers.
‡Accuracy not evaluated; checkpoint not preserved.

parameter budgets with identical GRPO configurations; the only variable is which structures were removed.

4.3 GRPO Responsiveness and Scale

Table 2 shows that GRPO accuracy changes are small across all pruned methods ($|\Delta| < 1$ pp: CKA-only +0.48, Standard +0.55, RASP −0.61 pp) despite the 4.4× peak reward gap between CKA-only and Standard. The unpruned model maintains 81.27% while reward rises to 0.669, confirming reward-accuracy decoupling under GSM8K's binary reward. This decoupling holds even at 25% sparsity, where Standard's peak reward collapses to 0.008 yet accuracy changes by only +0.99 pp (single seed; Appendix L). Extended training to 300 steps yields unchanged accuracy (CKA-only: 64.37%→64.37%, Standard: 59.44%→59.29%; single seed), confirming that reward gains do not translate to accuracy improvements within our training horizon. This decoupling extends across MATH-500 (Hendrycks et al., 2021), HumanEval (Chen et al., 2021), and MBPP (Austin et al., 2021) (all GRPO deltas < 2 pp; Appendices F, G), while the pruning-criterion ranking transfers to MATH-500 (CKA-only 29.0% vs Standard 26.4% vs RASP 22.6%). ShortGPT (Men et al., 2024) depth pruning shows task-dependent GRPO interaction (−1.36 pp on GSM8K, +4.01 pp on MBPP; Tables 2, 8), unlike the uniformly neutral deltas of fine-grained methods ($|\Delta| \leq 1.34$ pp).

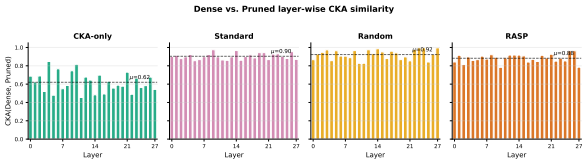


Figure 4: Layer-wise CKA similarity between dense and pruned representations (28 layers). CKA-only diverges most from dense (μ=0.62) yet recovers best; Standard (μ=0.90), Random (μ=0.92), and RASP (μ=0.88) preserve dense fidelity but collapse under RL.

4.4 Additional Results

Post-SFT accuracy does not predict GRPO outcome. Under SFT alone, CKA-only (63.91%) > RASP (61.49%) > Standard (59.29%), yet under GRPO the RASP/Standard ordering reverses (peak 0.081 vs 0.061). More strikingly, ShortGPT post-SFT (44.73%) shows GRPO actively harming accuracy (−1.36 pp), while unstructured pruning achieves the highest post-SFT accuracy (87.87%) with GRPO accuracy 87.64% (Δ=−0.23 pp), confirming that pruning criteria are irrelevant when structural damage is minimal (96.5% dense retention). Post-SFT accuracy does not monotonically determine GRPO recovery outcome; the recovery capacity gradient is driven by structural properties of the pruned architecture, not residual SFT quality.

Additional results. All methods collapse at 30% sparsity, validating 20% as the operating point (Appendix J). Pruning reduces parameters by 19% and memory by ∼28% (Appendix C). The CKA coefficient α shows a non-monotonic pattern across $\{0.0, 0.25, 0.5, 1.0, 2.0, 4.0\}$ (Appendix D).

4.5 Extended Analysis

Representational diversity. CKA similarity between dense and pruned representations (Figure 4) shows CKA-only diverges most from the dense model (μ=0.62), while Standard (μ=0.90), Ran-

Table 3: Cross-scale validation at 20% sparsity. 7B: DeepSeek-R1-Distill-Qwen-7B (GSM8K mean over 3 seeds for CKA-only and Standard; see Table 2). 1.5B: DeepSeek-R1-Distill-Qwen-1.5B (single seed). GSM8K accuracy (N=1,319). 7B reward = peak across 100 GRPO steps (cf. Table 1); 1.5B reward = mean across training steps.

Method	7B		1.5B	
	Peak Rew.	GSM8K (%)	Rew.	GSM8K (%)
CKA-only	**0.354**	**64.39**	**0.238**	**35.48**
Standard	0.081	59.84	0.084	22.29
RASP	0.061	60.88	0.099	26.16
Random	0.190	–	–	–

dom (μ=0.92), and RASP (μ=0.88) cluster near 1.0. Gradient importance preserves dense fidelity at the cost of redundant representations that provide no exploration benefit during GRPO.

Extended training cannot rescue collapse. Extending GRPO from 100 to 300 steps (Figure 5, Appendix E), Standard shows zero net improvement (reward 0.054–0.058, $f_0 > 0.5$ throughout), while CKA-only sustains a 5.5× gap (0.308 vs 0.056 at step 300). The gap is structural, not a matter of training budget.

Cross-scale and cross-architecture generalization. On DeepSeek-R1-Distill-Qwen-1.5B at 20% sparsity (Table 3), the three-method ordering is preserved (CKA-only 35.48% ≫ RASP 26.16% > Standard 22.29%). To test cross-architecture generalization, we apply CKA-only and Standard pruning to Qwen2.5-Math-7B (Yang et al., 2024) at 20% sparsity (Table 13, Appendix M). CKA-only causes near-total initial damage (1.0% post-prune accuracy vs Standard's 18.5%), yet SFT recovers CKA-only to 51.50% ± 3.53 while Standard remains at 21.00% ± 2.88 (N=200). This post-prune accuracy reversal—where the method with lower initial accuracy achieves dramatically higher recovery—provides additional evidence that recoverability is shaped by preserved network structure, not starting accuracy. Cross-architecture validation covers the SFT recovery stage; GRPO validation on Qwen2.5-Math-7B is deferred to future work. The CKA-only advantage widens at lower sparsity: the accuracy gap grows from 4.55 pp at 20% to 7.20 pp at 15% (3-seed means; a 58% increase), with CKA-only at 15% retaining 80.1% of original dense performance (Appendix L).

5 Discussion and Conclusion

We present a controlled comparison demonstrating that the structural pruning criterion shapes post-RLVR recovery capacity. Under identical GRPO training at 20% sparsity, CKA-only pruning enables active learning (peak reward 0.354, 4.4× Standard), while importance-based methods collapse. Extended training to 300 steps confirms this gap is structural, not a matter of training budget.

A plausible mechanism is preserved representational diversity: CKA-only diverges most from the dense representation (μ=0.62 diagonal CKA) yet recovers best, while collapsing methods preserve dense fidelity ($\mu \geq 0.88$). Gradient importance removes structures that are representationally unique, reducing the exploration capacity available for RL. Cross-architecture experiments on Qwen2.5-Math-7B show a striking post-prune accuracy reversal (CKA-only 1.0%→51.50% vs Standard 18.5%→21.00% after SFT), and cross-scale experiments on 1.5B preserve the three-method ordering, providing initial evidence for the generality of the finding. The recovery gap widens at lower sparsity (4.55 pp at 20% to 7.20 pp at 15%, both 3-seed means; p<0.001), though this trend is based on two sparsity levels. Reward-accuracy decoupling is negligible across all four benchmarks (GSM8K, MATH-500, HumanEval, MBPP; all GRPO deltas < 2 pp), while the recovery capacity gradient persists, confirming recovery capacity as the appropriate diagnostic. A depth pruning baseline (ShortGPT) shows task-dependent GRPO interaction (−1.36 pp on GSM8K, +4.01 pp on MBPP), unlike the uniformly neutral deltas of fine-grained methods. Recovery-aware pruning criteria become critical in the structured pruning regime, where removing entire architectural components causes functional damage that RLVR alone cannot recover. At 20% unstructured sparsity, the model retains 96.5% accuracy—pruning criteria are irrelevant because there is little to recover. Structured pruning at the same sparsity drops to 60–64%, creating the regime where the choice of pruning criterion shapes recoverability. We note that Ji et al. (2025) report 95%+ recovery using knowledge distillation on a different model and compression method; the conditions differ sufficiently (distillation vs. RLVR, different base models) that direct numerical comparison is not meaningful. Our contribution is the *comparative* finding across pruning criteria, not the absolute recovery rate. Code, pruning masks, and

model checkpoints will be released upon acceptance.

Limitations

Architecture sensitivity. Our primary results are on DeepSeek-R1-Distill-Qwen-7B (DeepSeek-AI, 2025) with GQA (G=7), with cross-architecture validation on Qwen2.5-Math-7B limited to SFT recovery (N=200; Appendix M). The intra-layer CKA overlap dynamics may differ substantially in architectures with more KV heads, different MLP structures, or non-GQA attention (MHA, MQA). Whether the gradient generalizes to other model families (e.g., LLaMA, Mistral) and scales (1B–70B) remains an open question.

Benchmark scope and sparsity ratio. We evaluate on four benchmarks (GSM8K, MATH-500, HumanEval, MBPP) at a fixed 20% structured sparsity. While these cover math reasoning and code generation, they do not address logical, commonsense, or multi-modal reasoning. Cross-scale validation on 1.5B is limited to GSM8K: the 1.5B model produces near-zero accuracy on HumanEval (0–0.61%) and MBPP (0–0.80%) across all pruning methods, providing no discriminative signal for code tasks at this scale. The interaction between functional overlap and recoverability may behave differently at higher sparsity ratios where more structures must be removed and functional redistribution becomes harder.

Limited hardware configurations. We report inference latency on a single GPU type (RTX PRO 6000) with greedy decoding. Realized speedups depend on hardware, batch size, and framework-level kernel support for non-standard hidden dimensions; our measurements may not transfer to other deployment settings.

Seed diversity. CKA-only and Standard are validated over 3 GRPO seeds (Table 2); RASP and Random each use a single seed (N=1). Pruning calibration and SFT configurations are identical across all methods. The single-seed RASP and Random results should be interpreted with caution; additional replication would strengthen confidence in the reported effect sizes.

Ethics Statement

This work uses publicly available models (DeepSeek-R1-Distill-Qwen, Qwen2.5-Math) and established benchmarks (GSM8K, MATH-500, HumanEval, MBPP); no private data or human subjects are involved. Structured pruning reduces model size and inference memory, which can lower the computational cost and energy consumption of deploying large language models. Our research focuses on the technical analysis of pruning criteria and their interaction with reinforcement learning recovery; we do not foresee direct negative societal impacts from this work.

References

Joshua Ainslie, James Lee-Thorp, Michiel de Jong, Yury Zemlyanskiy, Federico Lebrón, and Sumit Sanghai. 2023. GQA: Training generalized multi-query transformer models from multi-head checkpoints. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*.

Saleh Ashkboos, Maximilian L. Croci, Marcelo Gennari do Nascimento, Torsten Hoefler, and James Hensman. 2024. SliceGPT: Compress large language models by deleting rows and columns. In *Proceedings of the 12th International Conference on Learning Representations (ICLR)*.

Jacob Austin, Augustus Odena, Maxwell Nye, Maarten Bosma, Henryk Michalewski, David Dohan, Ellen Jiang, Carrie Cai, Michael Terry, Quoc Le, and Charles Sutton. 2021. Program synthesis with large language models. *arXiv preprint arXiv:2108.07732*.

Mark Chen, Jerry Tworek, Heewoo Jun, Qiming Yuan, Henrique Ponde de Oliveira Pinto, Jared Kaplan, Harri Edwards, Yuri Burda, Nicholas Joseph, Greg Brockman, and 1 others. 2021. Evaluating large language models trained on code. *arXiv preprint arXiv:2107.03374*.

Xiaodong Chen, Yuxuan Hu, Xiaokang Zhang, Yanling Wang, Cuiping Li, Hong Chen, and Jing Zhang. 2025. P^2 law: Scaling law for post-training after model pruning. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL)*.

Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. 2021. Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*.

MohammadReza Davari, Stefan Horoi, Amine Natik, Guillaume Lajoie, Guy Wolf, and Eugene Belilovsky. 2023. Reliability of CKA as a similarity measure in deep learning.

DeepSeek-AI. 2025. Deepseek-r1: Incentivizing reasoning capability in llms via reinforcement learning. *arXiv preprint arXiv:2501.12948*.

Longwei Ding, Anhao Zhao, Fanghua Ye, Ziyang Chen, and Xiaoyu Shen. 2026. From LLMs to LRMs: Rethinking pruning for reasoning-centric models. *arXiv preprint arXiv:2601.18091*.

Yihong Dong, Xue Jiang, Yongding Tao, Huanyu Liu, Kechi Zhang, Lili Mou, Rongyu Cao, Yingwei Ma, Jue Chen, Binhua Li, Zhi Jin, Fei Huang, Yongbin Li, and Ge Li. 2025. RL-PLUS: Countering capability boundary collapse of LLMs in reinforcement learning with hybrid-policy optimization. *arXiv preprint arXiv:2508.00222*.

Zijian Feng, Hanzhang Zhou, Zixiao Zhu, Tianjiao Li, Jia Jim Deryl Chua, Lee Onn Mak, Gee Wah Ng, and Kezhi Mao. 2025. RestoreLCC: Restoring pruned large language models via lost component compensation. In *Advances in Neural Information Processing Systems (NeurIPS)*. Spotlight.

Jonathan Frankle and Michael Carbin. 2019. The lottery ticket hypothesis: Finding sparse, trainable neural networks. In *Proceedings of the 7th International Conference on Learning Representations (ICLR)*.

Leo Gao, John Schulman, and Jacob Hilton. 2023. Scaling laws for reward model overoptimization. In *Proceedings of the 40th International Conference on Machine Learning (ICML)*.

Bowei He, Lihao Yin, Hui-Ling Zhen, Xiaokun Zhang, Mingxuan Yuan, and Chen Ma. 2025. Paser: Post-training data selection for efficient pruned large language model recovery. *arXiv preprint arXiv:2502.12594*. Under review.

Dan Hendrycks, Collin Burns, Saurav Kadavath, Akul Arora, Steven Basart, Eric Tang, Dawn Song, and Jacob Steinhardt. 2021. Measuring mathematical problem solving with the MATH dataset. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Seungbeom Hu, Chanjun Park, Andrew Ferraiuolo, Sang-Ki Ko, Jinwoo Kim, Haein Song, and Jieung Kim. 2024. Mpruner: Optimizing neural network size with cka-based mutual information pruning. *arXiv preprint arXiv:2408.13482*.

Yixin Ji, Zhimin Lin, Juntao Li, Kaixin Wu, Kai Yao, Xu Jia, Linjian Mo, and Min Zhang. 2025. Self-healing: Recovering pruned large reasoning models via reinforcement learning. Under review.

Jaehoon Kim and Dongha Lee. 2026. OPSD compresses what RLVR teaches: A post-RL compaction stage for reasoning models. *arXiv preprint arXiv:2605.06188*.

Max Klabunde, Tobias Schumacher, Markus Strohmaier, and Florian Lemmerich. 2023. Similarity of neural network models: A survey of functional and representational measures. *ACM Computing Surveys*, 57(9).

Steven Kolawole, Lucio Dery, Jean-François Kagy, Virginia Smith, Graham Neubig, and Ameet Talwalkar. 2025. Everybody prune now: Structured pruning of llms with only forward passes. In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*.

Simon Kornblith, Mohammad Norouzi, Honglak Lee, and Geoffrey Hinton. 2019. Similarity of neural network representations revisited. In *Proceedings of the 36th International Conference on Machine Learning (ICML)*.

Alex Lachance. 2022. Using centered kernel alignment for understanding and pruning neural networks. Master's thesis, University of Rhode Island.

Yiheng Liu, Junhao Ning, Sichen Xia, Xiaohui Gao, Ning Qiang, Bao Ge, Junwei Han, and Xintao Hu. 2025a. Pruning large language models by identifying and preserving functional networks. *arXiv preprint arXiv:2508.05239*.

Yijiang Liu, Huanrui Yang, Youxin Chen, Rongyu Zhang, Miao Wang, Yuan Du, and Li Du. 2025b. PAT: Pruning-aware tuning for large language models. In *Proceedings of the 39th AAAI Conference on Artificial Intelligence (AAAI)*.

Ryan Lucas, Kayhan Behdin, Zhipeng Wang, Qingquan Song, Shao Tang, and Rahul Mazumder. 2025. Reasoning models can be accurately pruned via chain-of-thought reconstruction. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Xinyin Ma, Gongfan Fang, and Xinchao Wang. 2023. Llm-pruner: On the structural pruning of large language models. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Kohsei Matsutani, Shota Takashiro, Gouki Minegishi, Takeshi Kojima, Yusuke Iwasawa, and Yutaka Matsuo. 2026. Rl squeezes, sft expands: A comparative study of reasoning llms. In *Proceedings of the 14th International Conference on Learning Representations (ICLR)*.

Xin Men, Mingyu Xu, Qingyu Zhang, Bingning Wang, Hongyu Lin, Yaojie Lu, Xianpei Han, and Weipeng Chen. 2024. ShortGPT: Layers in large language models are more redundant than you expect. *arXiv preprint arXiv:2403.03853*.

Ocean Monjur, Shahriar Kabir Nahin, and Anshuman Chhabra. 2026. Doing more with less: Revisiting the effectiveness of LLM pruning for test-time scaling. *arXiv preprint arXiv:2604.25098*.

Vedant Shah, Johan Obando-Ceron, Vineet Jain, Brian Bartoldson, Bhavya Kailkhura, Sarthak Mittal, Glen Berseth, Pablo Samuel Castro, Yoshua Bengio, Nikolay Malkin, Moksh Jain, Siddarth Venkatraman, and Aaron Courville. 2025. A comedy of estimators: On KL regularization in RL training of LLMs. *arXiv preprint arXiv:2512.21852*.

Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Mingchuan Zhang, Y.K. Li, Y. Wu, and Daya Guo. 2024. Deepseekmath: Pushing the

limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300*.

Mingjie Sun, Zhuang Liu, Anna Bair, and J. Zico Kolter. 2024. A simple and effective pruning approach for large language models. In *Proceedings of the 12th International Conference on Learning Representations (ICLR)*.

Ziyan Wang, Enmao Diao, Qi Le, Pu Wang, Guanchu Wang, Minwoo Lee, Shu-ping Yeh, and Li Yang. 2025. Think before you prune: Self-reflective structured pruning for reasoning language models. *arXiv preprint arXiv:2512.02185*.

Ziyang Wang, Jiangfeng Xiao, Chuan Xiao, Ruoxiang Li, Rui Mao, and Jianbin Qin. 2026. Grasprune: Global gating for budgeted structured pruning of large language models. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (ACL)*.

Yang Xiang, Yixin Ji, Juntao Li, and Min Zhang. 2025. Think before you prune: Selective self-generated calibration for pruning large reasoning models. *arXiv preprint arXiv:2511.18864*.

An Yang, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chengyuan Li, and 1 others. 2024. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*.

Tao Yu, Yongqi An, Kuan Zhu, Guibo Zhu, Ming Tang, and Jinqiao Wang. 2025. FANG: Improving generalization in LLM structured pruning via function-aware neuron grouping. *arXiv preprint arXiv:2512.23014*.

Yiran Zhao, Shengyang Zhou, Zijian Wu, Tongyan Hu, Yuhui Xu, Rengan Dou, Kenji Kawaguchi, Shafiq Joty, Junnan Li, and Michael Qizhe Shieh. 2026. Gradually compacting large language models for reasoning like a boiling frog. *arXiv preprint arXiv:2602.04919*.

A Training Details

Table 4: Hyperparameter details for SFT warmup and GRPO recovery stages.

Parameter	SFT	GRPO
Learning rate	2×10^{-5}	1×10^{-6}
Batch size	1	1
Gradient accumulation	8	1
Max sequence length	2,048	–
Max new tokens	–	1,024
Optimizer	AdamW	AdamW
Epochs / Steps	2 epochs	100 steps
KL coefficient β	–	0.01
Sampling temperature	–	0.4
Num generations per prompt	–	8
Reward	–	Binary (exact match)
Loss formulation	Cross-entropy	DAPO-style clipped
`reward_desert_patience`	–	50

B Algorithm

Algorithm 1 Controlled Pruning Comparison Framework

Require: Model $\mathcal{M}$, calibration data $\mathcal{D}_{\text{cal}}$, target sparsity s, scoring criterion $\mathcal{C} \in \{\text{Std}, \text{RASP}, \text{CKA}, \text{Rand}\}$, layer floor τ

Ensure: Pruned and recovered model $\mathcal{M}'$

% Stage 1: Shared Structural Analysis

for each structure $h \in \mathcal{H}$ **do**

  Compute importance $I(h)$ via Eq. (1)

  Collect activations $\mathbf{X}_h$ on $\mathcal{D}_{\text{cal}}$

for each layer $\ell = 1, \ldots, L$ **do**

  for each structure h_i in layer ℓ **do**

    $F(h_i) \leftarrow \frac{1}{|\mathcal{H}_\ell|-1} \sum_{j \neq i} \text{CKA}(\mathbf{X}_{h_i}, \mathbf{X}_{h_j})$

% Stage 2: Criterion-Specific Scoring

for each structure $h \in \mathcal{H}$ **do**

  $S(h) \leftarrow \mathcal{C}(I(h), F(h))$ ▷ Eq. 4, 5, 6, or random

Sort $\mathcal{H}$ by $S(h)$ ascending; prune lowest up to s, subject to τ

% Stage 3: Identical Two-Stage Recovery

$\mathcal{M}_{\text{warm}} \leftarrow \text{SFT}(\mathcal{M}_{\text{pruned}}, \mathcal{D}_{\text{sft}})$ ▷ restore EOS & format

$\mathcal{M}' \leftarrow \text{GRPO}(\mathcal{M}_{\text{warm}}, \mathcal{D}_{\text{rlvr}})$ ▷ recover reasoning

return $\mathcal{M}'$

C Inference Efficiency

Table 5: Inference characteristics at 20% sparsity (clean v2 benchmark, single-GPU, no contention). Pruning reduces parameters by 19% and memory by ∼28%, with TTFT improving 18%. Per-token throughput decreases 19% (0.81×) due to non-uniform layer widths from full-layer removal that current GPU kernels do not optimize.

Configuration	TPS (tok/s)	TTFT (ms)	Per-token (ms)	Params (B)
Dense 7B	72.0	20.1	13.9	7.62
CKA-only 20%	58.3	16.4	16.7	6.16
Standard 20%	58.2	16.4	16.7	6.16

All pruned models achieve ∼28% memory reduction from structural parameter removal. TTFT (time to first token) improves by 18% (20.1→16.4 ms), reflecting faster prefill with fewer parameters. Per-token throughput decreases by 19% (72.0→58.3 tok/s, or 13.9→16.7 ms per token), consistent with non-uniform layer widths creating computational bottlenecks that current GPU

Table 6: Effect of CKA coefficient α on post-prune accuracy at 20% sparsity (N=100). α=0.0 reduces to Standard importance-only pruning. Best in **bold**, second-best underlined.

α	Post-Prune Acc. (%)
0.00	27
0.25	23
0.50	22
1.00	31
2.00	<u>36</u>
4.00	**40**

kernels do not optimize away. CKA-only and Standard show identical throughput (58.3 vs 58.2 tok/s), confirming that the pruning criterion affects recovery capacity but not inference speed at matched sparsity.

D CKA Coefficient α Ablation

E Extended Training Dynamics

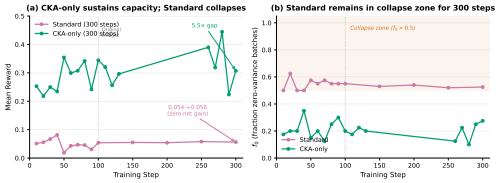


Figure 5: Extended GRPO training (300 steps, 3× original budget). (a) Standard reward shows zero net gain (0.054→0.056); CKA-only oscillates in the 0.22–0.44 range with a 5.5× gap at step 300. (b) Standard f_0 remains in the collapse zone (>0.5); CKA-only stays below 0.30.

F MATH-500 Out-of-Distribution Evaluation

G MBPP Code Generation

H CKA Scoring Stability

CKA-based structure rankings are highly stable across calibration data volumes: Spearman ρ between rankings from 256, 512, and 1024 calibration samples ranges from 0.964 to 0.979, indicating that the scoring criterion is not sensitive to calibration set size.

I Seed Reproducibility

CKA-only GSM8K accuracy across 3 GRPO seeds: 64.37% (seed 42), 64.22% (seed 123), 64.58% (seed 456), yielding mean 64.39% ± 0.19 pp. Standard: mean 59.84% ± 0.38 pp across the same 3 seeds. See Table 11 for the full summary.

Table 7: Out-of-distribution evaluation on MATH-500 (N=500, greedy decoding, `max_new_tokens=2048`). The recovery capacity gradient persists: CKA-only maintains a ∼6 pp advantage over importance-based methods.

Method	Recovery	MATH-500 (%)
Dense	–	53.0
CKA-only	SFT	29.0
CKA-only	GRPO	29.2
Standard	SFT	25.8
Standard	GRPO	26.4
RASP	SFT	22.6
RASP	GRPO	24.0

Over half of Dense model responses are truncated at 2048 tokens, depressing absolute accuracy (73.6% at 4096 tokens vs 53.0% at 2048). All pruned models are evaluated under the same constraint for fair comparison.

Table 8: MBPP code generation accuracy (N=374, greedy decoding). All fine-grained pruning methods converge near a capability floor (∼75% code capability loss vs Dense), limiting differentiation (∼1 pp range, within stderr). ShortGPT shows a larger GRPO delta (+4.01 pp) but at the 95% CI boundary (stderr ≈1.55 pp).

Method	SFT (%)	GRPO (%)	Δ (pp)
Dense	–	53.48	–
CKA-only 20%	12.83	14.17	+1.34
Standard 20%	12.03	13.10	+1.07
RASP 20%	11.76	13.10	+1.34
ShortGPT 20%†	10.43	14.44	+4.01

†Depth pruning; removes entire layers.

J Sparsity Cliff at 30%

All tested methods collapse at 30% sparsity: Standard retains 4% accuracy, RASP and CKA-only reach 0%. This capacity threshold, independent of pruning criterion, validates 20% as the operating point and confirms that the recovery capacity gradient reflects how methods allocate a limited pruning budget.

K Multi-Seed GRPO Results

At 20% sparsity, the 4.55 pp gap between CKA-only and Standard is robust across seeds, with non-overlapping ranges (CKA-only: 64.22–64.58%, Standard: 59.44–60.20%). Standard shows higher

Table 9: CKA scoring stability across calibration sizes. Spearman ρ between structure rankings produced by different calibration set sizes. High correlation ($\rho \geq 0.964$) indicates that CKA-based scoring is robust to calibration data volume.

Calibration Sizes	Spearman ρ (min)	Spearman ρ (mean)
256 / 512 / 1024	0.964	0.979

Table 10: Sparsity cliff at 30% structured sparsity (N=100). All tested pruning criteria collapse, with near-zero accuracy and EOS generation rates below 6%.

Method	Acc. (%)	EOS Rate (%)
Standard	4	5.5
RASP	0	3.0
CKA-only	0	–[†]

[†]Terminated early after collapse confirmed.

seed variance (0.38 pp vs 0.19 pp), consistent with borderline-collapse dynamics where small perturbations produce larger outcome variation. Note: Standard seed 42 was trained from a separately initialized SFT checkpoint (accuracy 59.21% vs. 59.29% for seeds 123/456; $\Delta = 0.08$ pp). At 15% sparsity, the gap widens to 7.20 pp (72.76% vs 65.56%, $t \approx 16.4$, $p < 0.001$), with non-overlapping ranges (CKA-only: 72.10–73.09%, Standard: 65.13–65.43%).

L Cross-Sparsity Validation

At 15% sparsity (3-seed mean), the CKA-only advantage widens to +7.20 pp (72.76% vs 65.56%), compared to +4.55 pp at 20%. CKA-only at 15% (72.76%) represents the highest mean accuracy across all experimental conditions, retaining 80.1% of original dense model performance (90.83%). At 25% sparsity (single seed), Standard's SFT accuracy decline accelerates (−7.20 pp from 15%→20%, −8.42 pp from 20%→25%), while CKA-only's decline decelerates (−8.27 pp then −3.79 pp). Reward-accuracy decoupling is preserved ($|\Delta| \leq 1$ pp for both methods) despite Standard's near-zero peak reward (0.008).

M Cross-Architecture Validation

CKA-only pruning causes near-total initial damage on Qwen2.5-Math-7B (1.0% post-prune accuracy vs Standard's 18.5%), yet SFT recovers CKA-only to 51.50% while Standard remains at 21.00%. This post-prune accuracy reversal provides strong evidence that recovery capacity is shaped by preserved network structure rather than starting accuracy. EOS rates remain below 40% for both methods, indicating that SFT warmup is less effective on this architecture; GRPO recovery may yield further improvements but was not run due to compute constraints.

Table 11: Multi-seed GRPO results on GSM8K (N=1,319). CKA-only and Standard each trained with 3 GRPO seeds (42, 123, 456); pruning calibration and SFT are identical across all runs within each sparsity level.

Sparsity	Method	Mean Acc. (%)	Std (pp)	Seeds
20%	CKA-only	64.39	0.19	3
	Standard	59.84	0.38	3
	RASP	60.88	–	1
15%	CKA-only	72.76	0.57	3
	Standard	65.56	0.50	3

Table 12: Cross-sparsity validation on DeepSeek-R1-Distill-Qwen-7B. GSM8K accuracy (N=1,319) after SFT warmup and GRPO recovery. 15% and 20% GRPO values are mean over 3 seeds (see Tables 2, 11). [‡]Single seed. Mean $|\Delta| \leq 1$ pp across all conditions.

Sparsity	Method	SFT (%)	GRPO (%)	Δ (pp)
15%	CKA-only	72.18	**72.76**±**0.57**	+0.58
	Standard	66.49	65.56±0.50	−0.93
20%	CKA-only	63.91	64.39	+0.48
	Standard	59.29	59.84	+0.55
25%[‡]	CKA-only	60.12	59.59	−0.53
	Standard	50.87	51.86	+0.99
30%	All	Collapse (≤4%; Table 10)		

N KL Divergence During GRPO Training

Table 13: Cross-architecture validation on Qwen2.5-Math-7B (Yang et al., 2024) at 20% structured sparsity. Accuracy evaluated on GSM8K (N=200). Post-SFT reports mean ± std over multiple SFT runs. EOS rate: fraction of generations that produce a valid end-of-sequence token.

Method	Post-Prune (%)	Post-SFT (%)	EOS Rate (%)
CKA-only	1.0	**51.50 ± 3.53**	27.0
Standard	18.5	21.00 ± 2.88	36.5

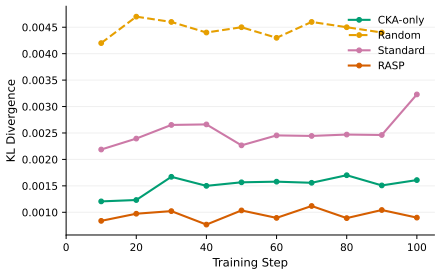


Figure 6: KL divergence between the policy and reference model during GRPO training. RASP maintains near-zero KL (<0.001), consistent with minimal policy movement during collapse. Random shows the highest KL (∼0.004–0.005), indicating more policy exploration. CKA-only and Standard fall between these extremes.